Technical Lesson: Secure Network Architecture
You are a newly hired Junior Cybersecurity Engineer at Acme Corp, a growing company that handles sensitive financial and human resources data. Recently, a security audit revealed a significant risk: all internal network devices are on a flat network, meaning that if an attacker gains access to one machine, they could easily move laterally and access critical systems across departments.

To address this risk, your manager has tasked you with implementing network segmentation to isolate the Finance and HR departments, reducing the risk of unauthorized access. Your goal is to configure a VyOS router to create two separate subnets, ensuring strict traffic isolation unless explicitly permitted. You will also validate the segmentation using tcpdump, a powerful network traffic analysis tool.

By the end of this lesson, you'll be able to set up a VyOS router to split your company's network into two secure zones: Finance and HR. You'll assign static IP addresses to each device so everything's in the right place, and then test your setup by running ping checks and watching traffic with tcpdump. Along the way, you'll learn why network segmentation is such a big deal for stopping attackers from moving around once they get inside. If something doesn't work as expected, you'll also learn how to troubleshoot and fix common network and security setup issues just like you would on the job.

Tools and Resources
Tools

- **VyOS Router:** **https://drive.google.com/file/d/18ujHjtooze9-BFuaeMojGV54EEoHxc5t/view?usp=drive_link**
- **Tcpdump:** A command-line packet capture tool, pre-installed on Ubuntu Server, used for traffic analysis and validation.
- **Ping Command:** A built-in tool on Ubuntu Server for basic connectivity testing between devices.

Resources

- **Ubuntu Server ISO (24.04):** These will act as the endpoints for each subnet. https://releases.ubuntu.com/24.04/
- https://docs.google.com/document/d/10Tlo4SaB8ONGgiNXQm_cu2k3y5A_mr_iftgAhbgu8T4/copy: **this will prompt you to make a copy.**

Instructions
Task 1. Download the Required Resources
Download the necessary files to set up your environment.

- Go to Ubuntu 24.04 Releases and download the server 64-bit PC (AMD64) install image.
- Access the VyOS OVA provided in the course materials.

Task 2. Set Up the Virtual Machines

This step involves creating two Ubuntu Server VMs, installing the OS, and configuring network settings.

Step 1: Create the Virtual Machine in VirtualBox

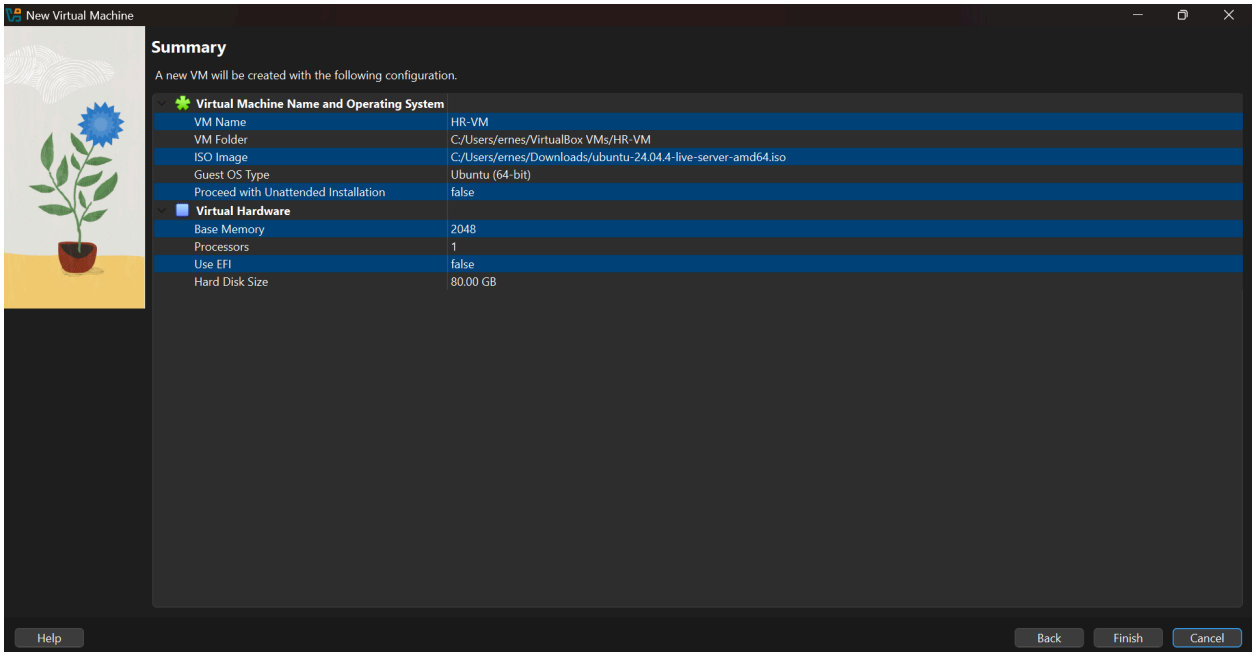
1. Open VirtualBox and click New to create a new VM.
2. Name the first VM Finance-VM (and the second VM HR-VM) and select the Ubuntu Server ISO (24.04) as the ISO image.

Step 2: Install Ubuntu Server

1. Set RAM to **2 GB** and **HDD** to **80 GB.**

Finance-VM

HR-VM



2. Keep the default Ubuntu Server
3. Edit the IPv4 in the Network Configuration and set the method to Manual:

Finance-VM:

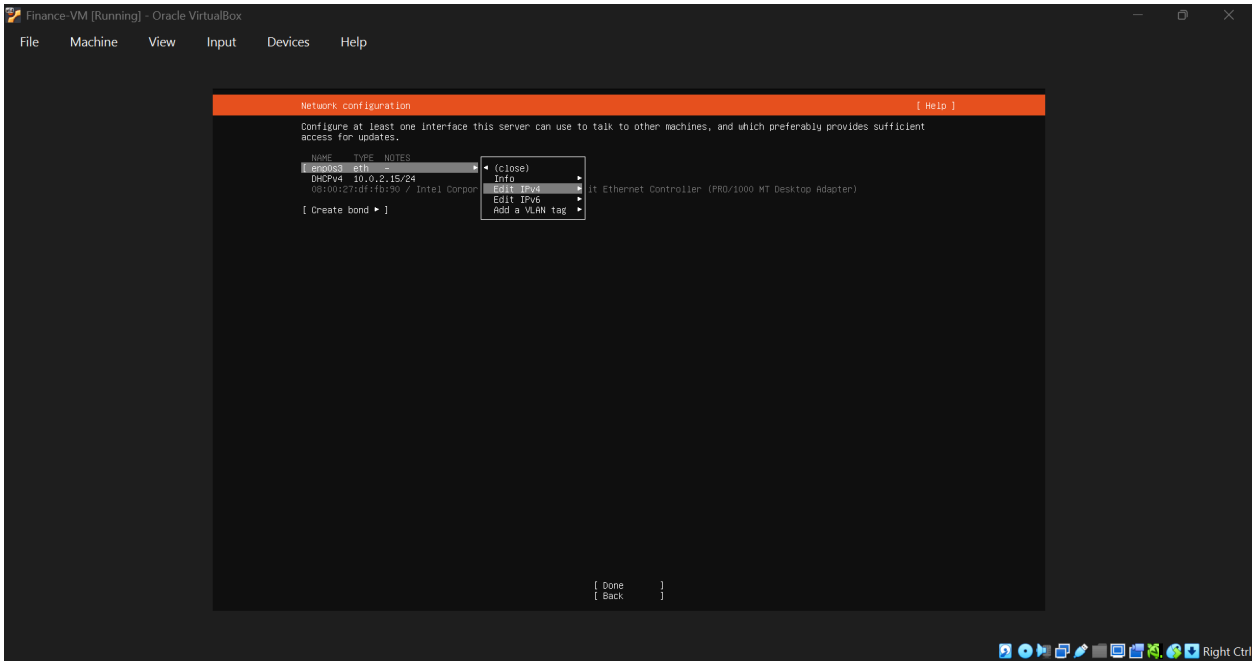
Subnet: 192.168.1.0/24

Address: 192.168.1.2

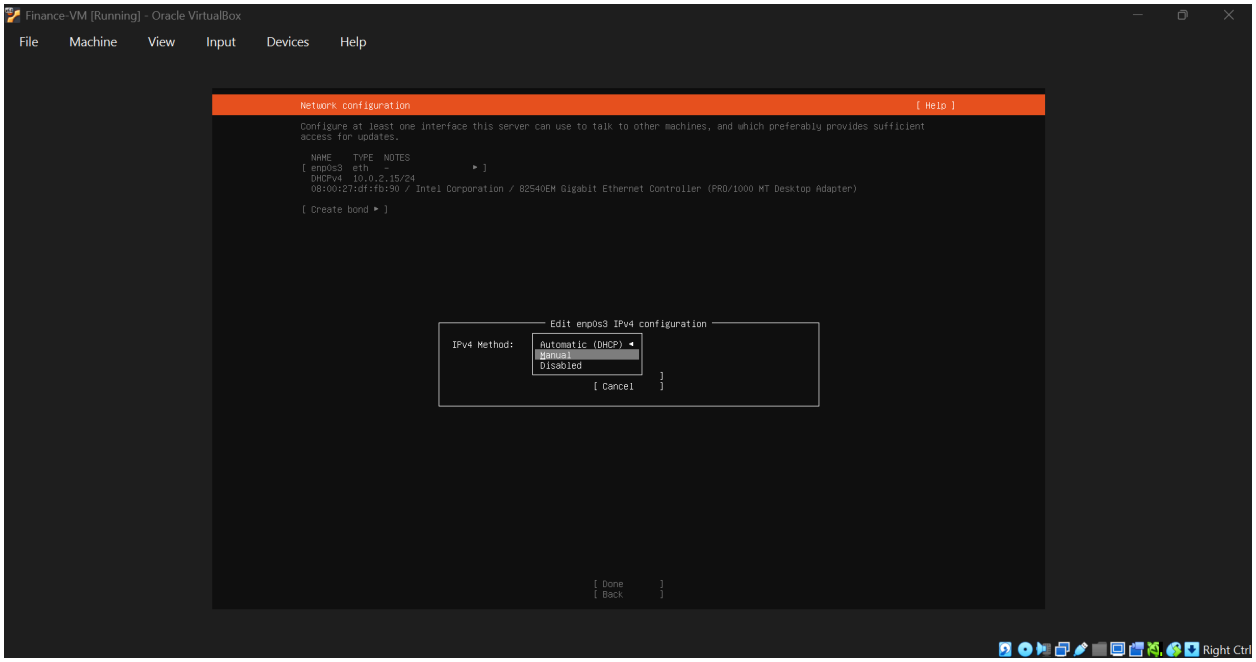
Gateway: 192.168.1.1

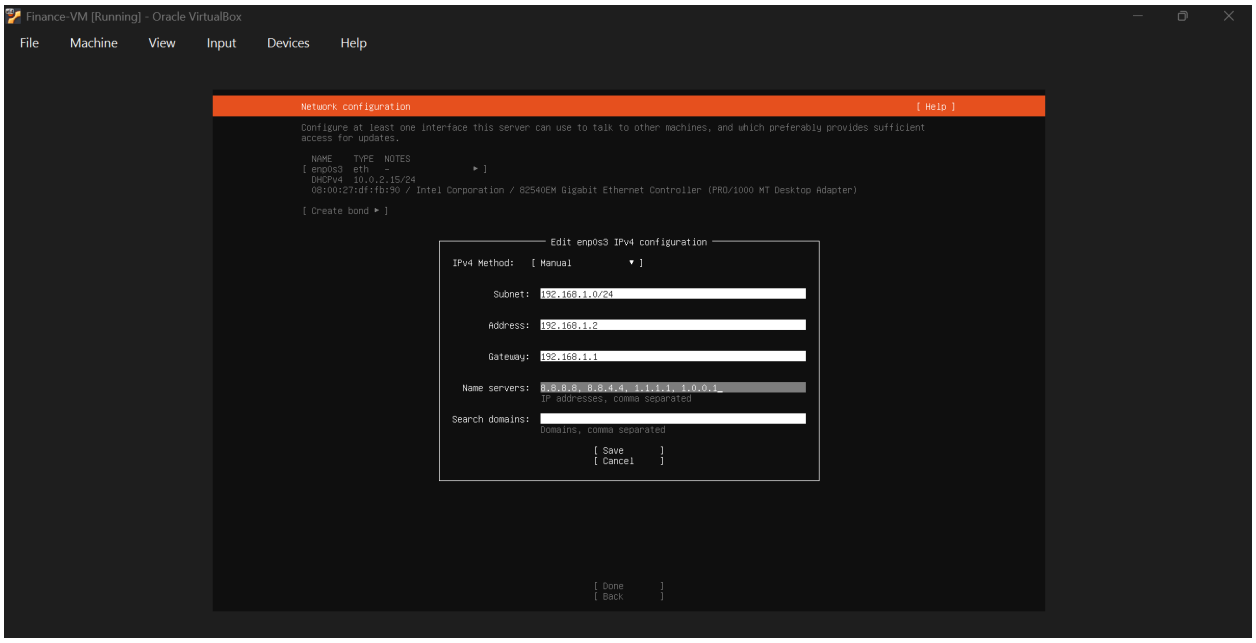
Edit the IPv4 in the Network Configuration and set the method to Manual as shown below.

Editing the IPv4



Setting the method to Manual


Finance-VM [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Network configuration
[Help]
Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.
NAME TYPE NOTES
[enp0s3 eth - ▸]
DHCPv4 10.0.2.15/24
08:00:27:df:fb:90 / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)
[Create bond ▸]
Edit enp0s3 IPv4 configuration
IPv4 Method:
Automatic (DHCP) ◂
Manual
Disabled
[Cancel]
[Done]
[Back]
Right Ctrl


Finance-VM [Running] - Oracle VirtualBox
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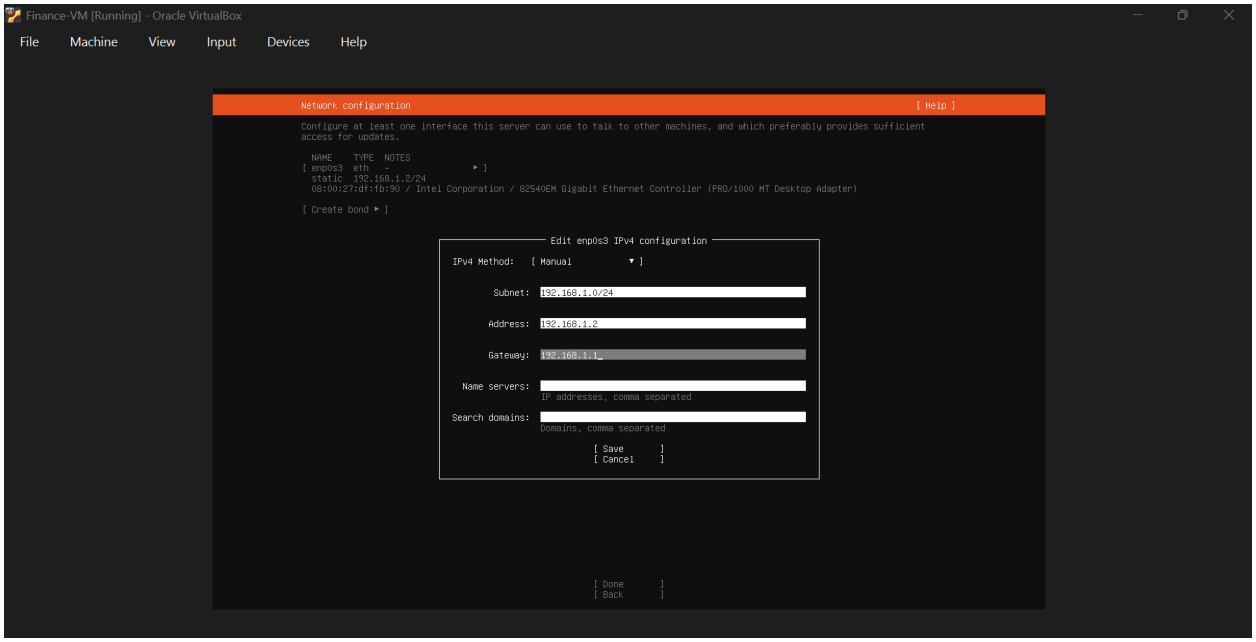
Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.
NAME TYPE NOTES
[enp0s3 eth - ▸]
DHCPv4 10.0.2.15/24
08:00:27:df:fb:90 / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)
[Create bond ▸]
Edit enp0s3 IPv4 configuration
IPv4 Method: [Manual ▾]
Subnet: 192.168.1.0/24
Address: 192.168.1.2
Gateway: 192.168.1.1
Name servers: 8.8.8.8, 8.8.4.4, 1.1.1.1, 1.0.0.1
IP addresses, comma separated
Search domains:
Domains, comma separated
[Save]
[Cancel]
[Done]
[Back]



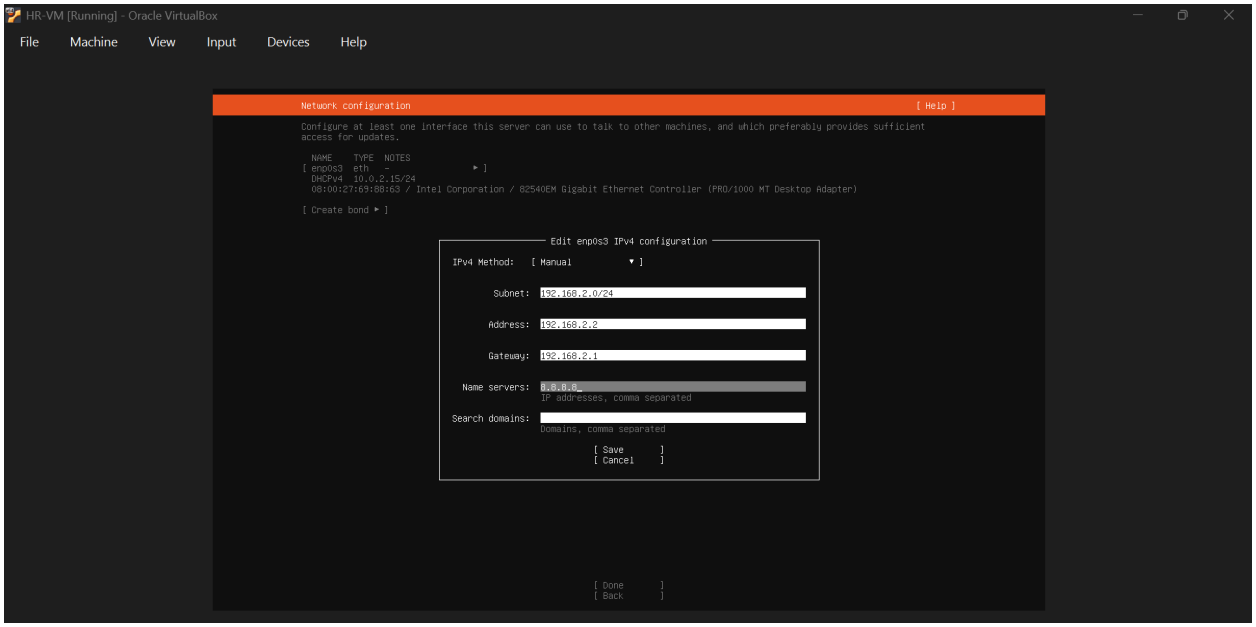
HR-VM:

Subnet: 192.168.2.0/24

Address: 192.168.2.2

Gateway: 192.168.2.1

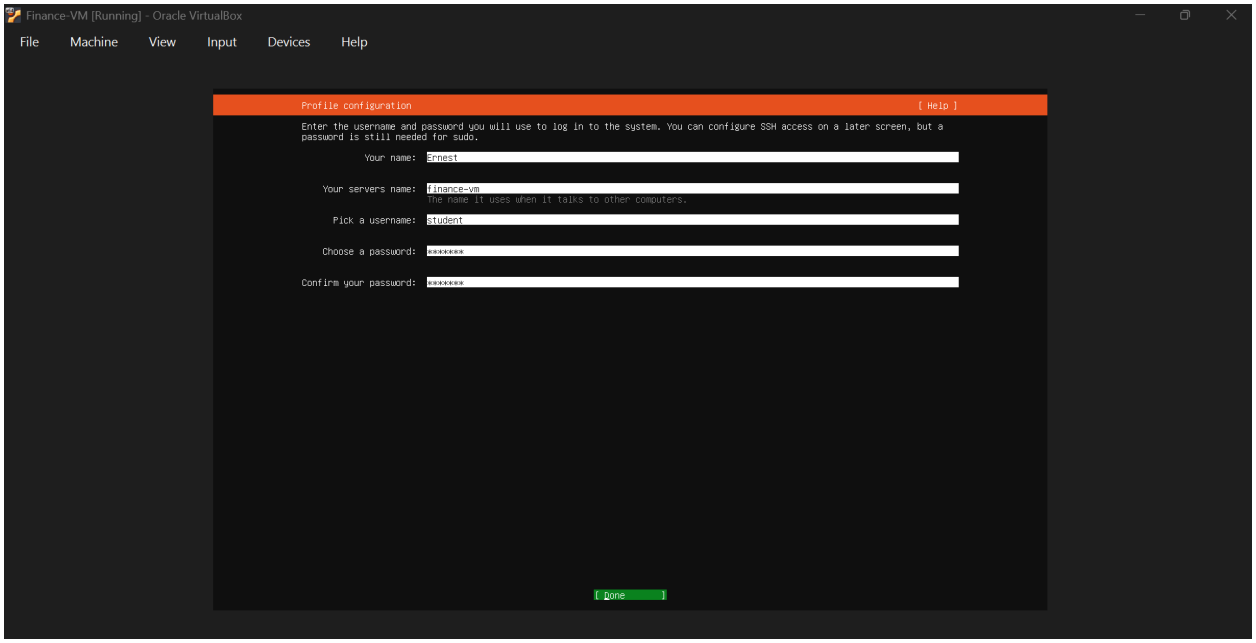
Solution:



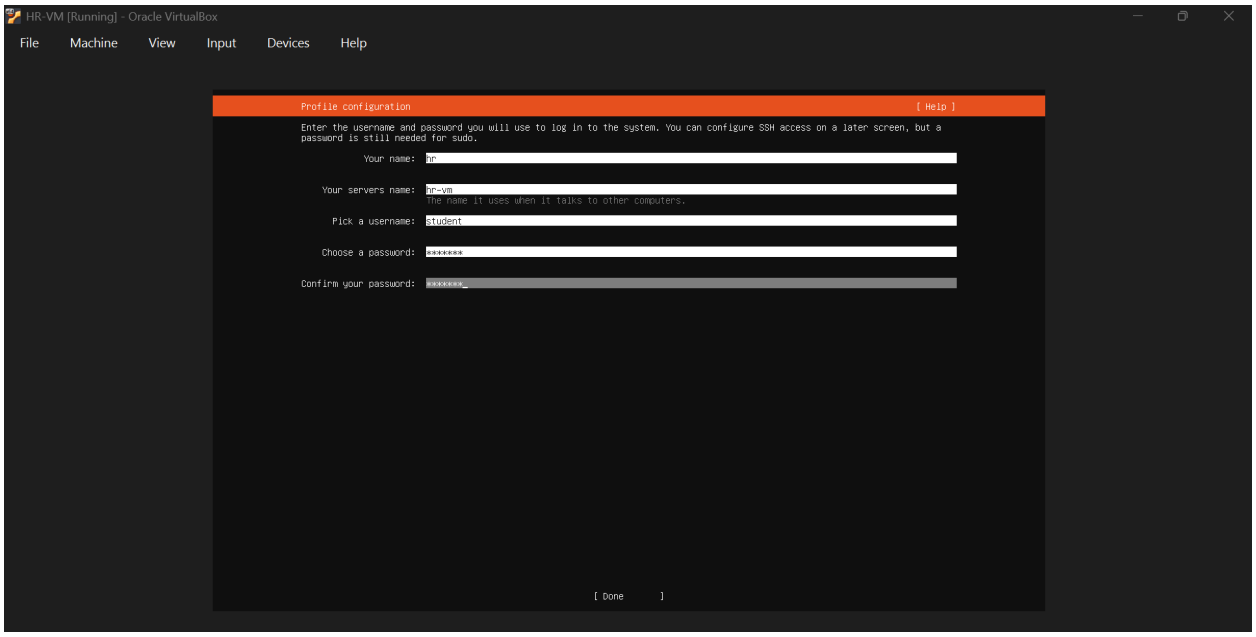
4. Leave the rest of the setup as the default until profile configuration:

Username & Password: student/student and the server's name to finance-vm/hr-vm.

Finance-vm



hr-vm



5. Skip upgrading to Ubuntu Pro and skip installing OpenSSH. It is not needed for this module. Skip any Snaps.
6. Continue installation and complete setup.

HR-VM [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Installing system

```
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
 curtin command in-target
installing system
 executing curtin install initial step
 executing curtin install partitioning step
  curtin command install
   configuring storage
    running 'curtin block-meta simple'
     curtin command block-meta
      removing previous storage devices
      configuring disk: disk-sda
     configuring partition: partition-0
     configuring partition: partition-1
     configuring format: format-0
     configuring partition: partition-2
     configuring lvm_volgroup: lvm_volgroup-0
     configuring lvm_partition: lvm_partition-0
     configuring format: format-1
     configuring mount: mount-1
     configuring mount: mount-0
 executing curtin install extract step
  curtin command install
   writing install sources to disk
    running 'curtin extract'
     curtin command extract
      acquiring and extracting image from cp:///tmp/tmplyawt_in/mount
 configuring keyboard
  curtin command in-target
 executing curtin install curthooks step
  curtin command install
   configuring installed system
    running 'curtin curthooks'
     curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      Installing packages on target system: ['grub-pc']
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel /
```

[View full log]

Finance-VM [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Installing system

```
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
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     curtin command block-meta
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      configuring disk: disk-sda
     configuring partition: partition-0
     configuring partition: partition-1
     configuring format: format-0
     configuring partition: partition-2
     configuring lvm_volgroup: lvm_volgroup-0
     configuring lvm_partition: lvm_partition-0
     configuring format: format-1
     configuring mount: mount-1
     configuring mount: mount-0
 executing curtin install extract step
  curtin command install
   writing install sources to disk
    running 'curtin extract'
     curtin command extract
      acquiring and extracting image from cp:///tmp/tmphfx9e0k3/mount
 configuring keyboard
  curtin command in-target
 executing curtin install curthooks step
  curtin command install
   configuring installed system
    running 'curtin curthooks'
     curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      Installing packages on target system: ['grub-pc']
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel |
```

[View full log]

Finance-VM [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
student@finance-vm:~$ _
```

HR-VM [Running] - Oracle VirtualBox

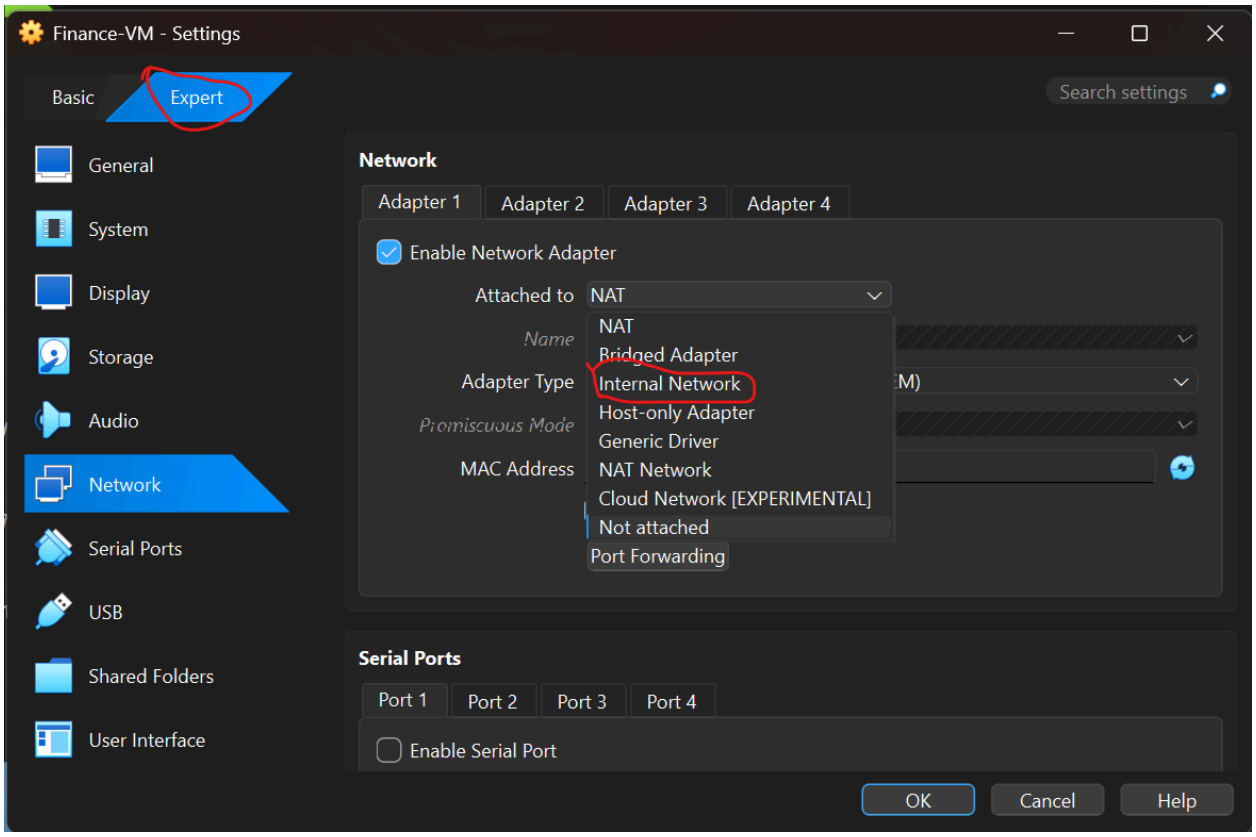
File Machine View Input Devices Help

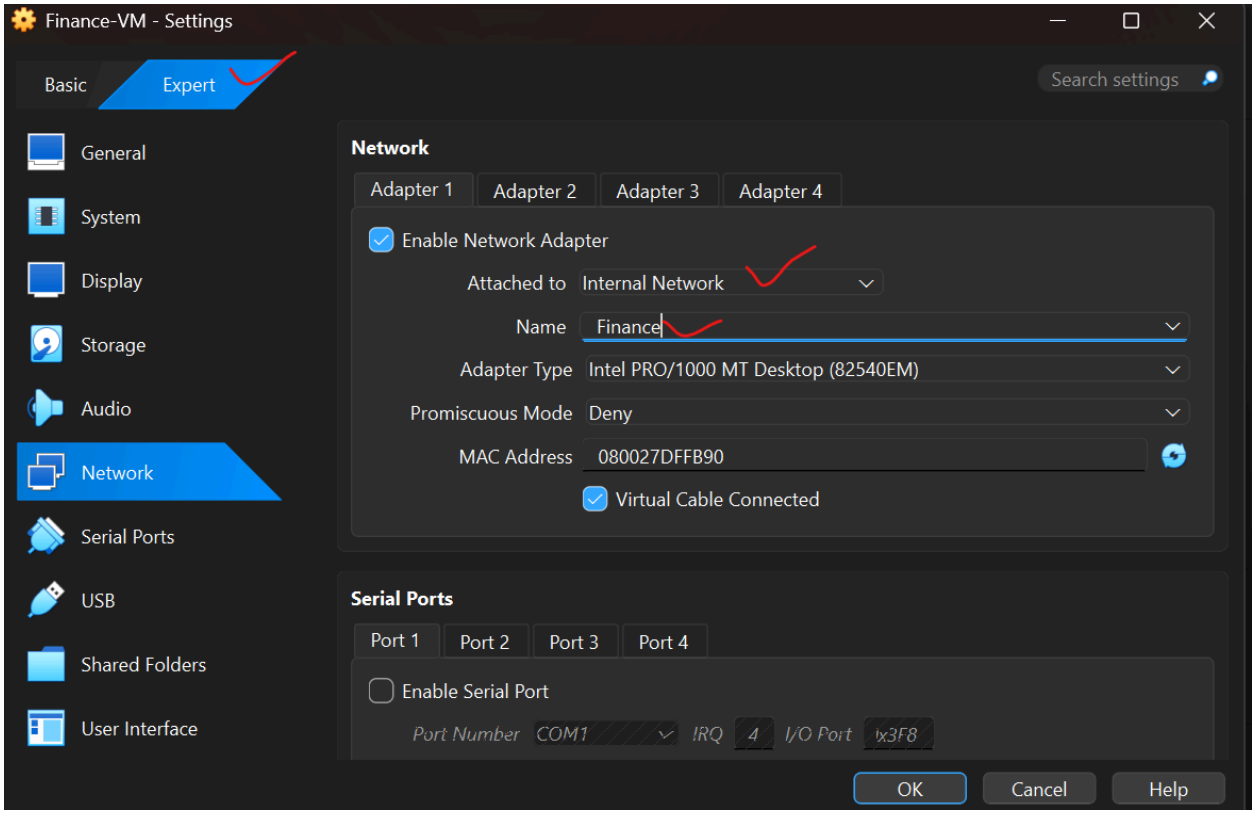
```
student@hr-vm:~$
```

Step 3: Change to Internal Networks in VirtualBox

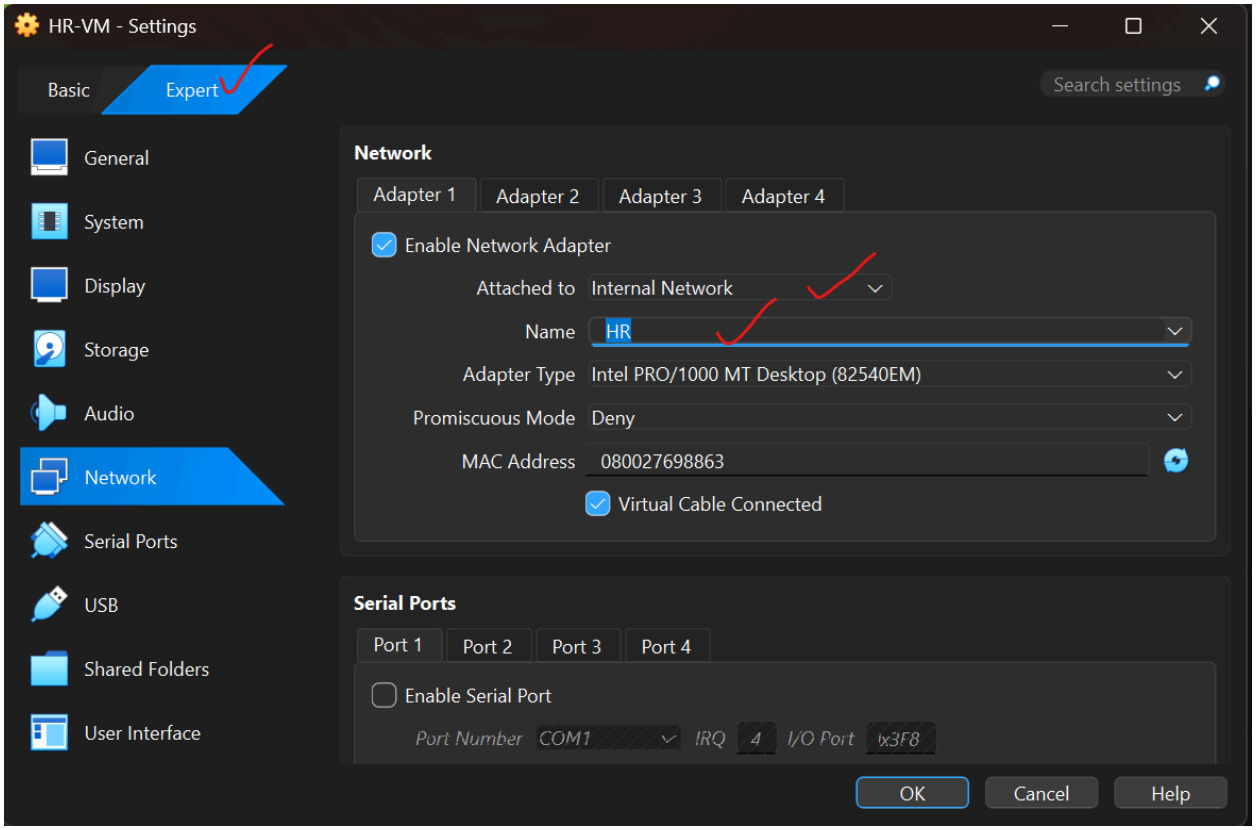
1. Shut down both VMs after installation is complete.
2. Open **VirtualBox** → **Settings** → **Network** for Finance-VM:

Adapter 1: Change from **NAT** to **Internal Network** and name it "**Finance**".





Open **VirtualBox** → **Settings** → **Network** for **HR-VM**:
Adapter 1: Change from **NAT** to **Internal Network** and name it "**HR**".



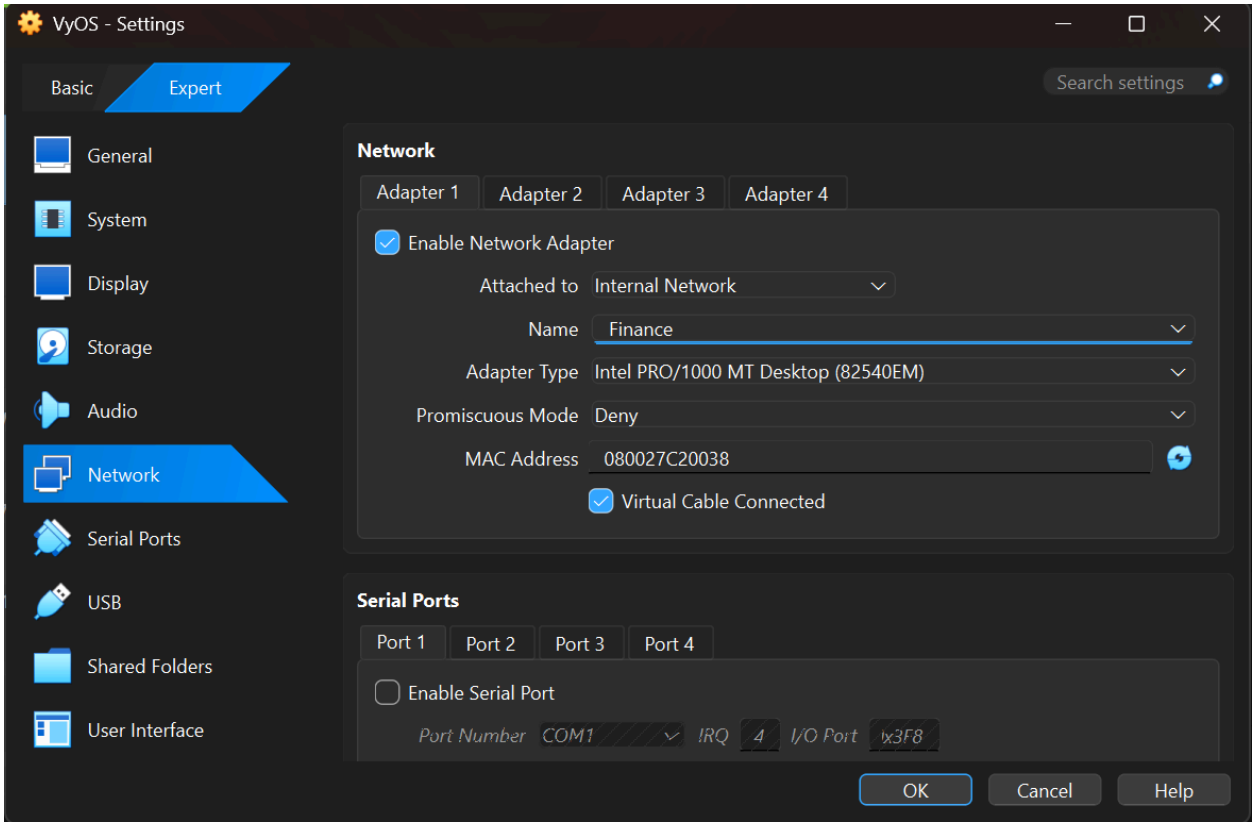
Start the VMs again.

Task 3. Configure the VyOS Router

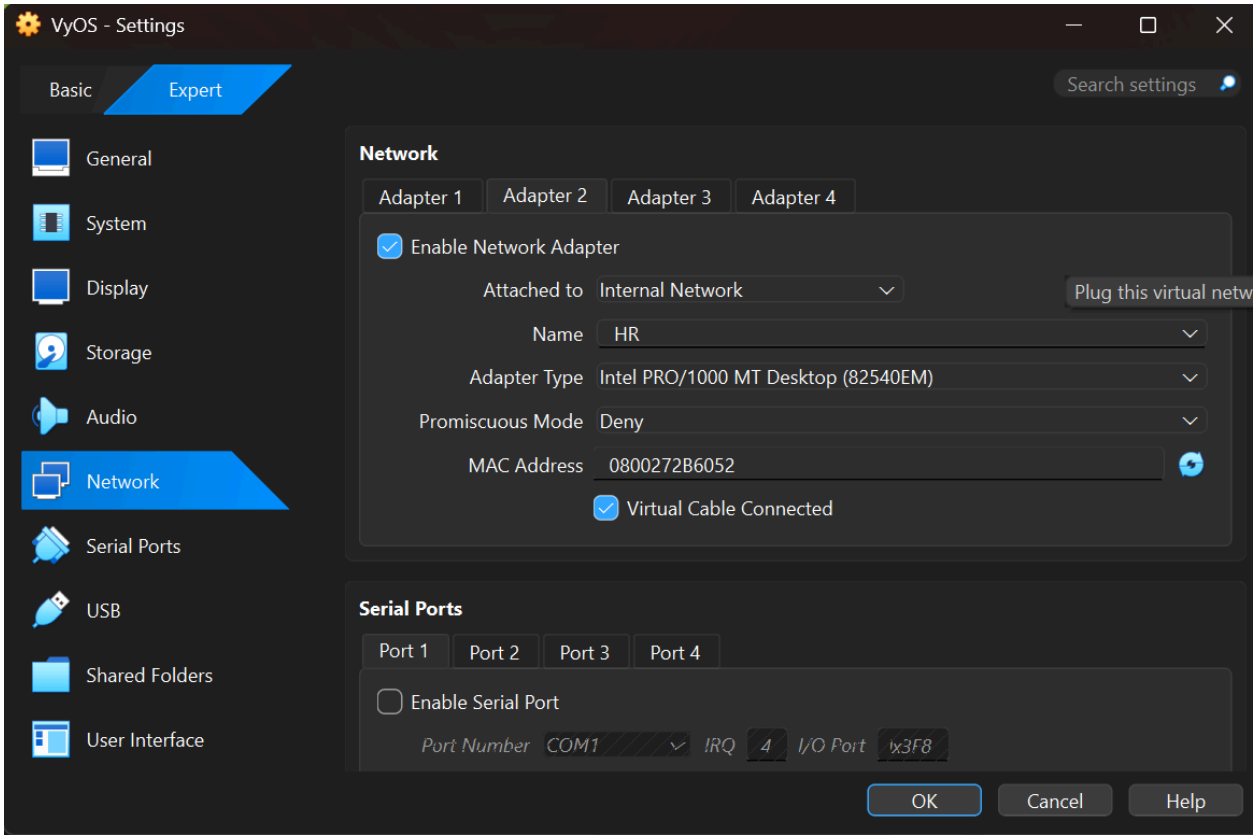
1. Import the VyOS OVA into your virtualization tool.

- Assign the Network **Adapter 1** to **Finance** and **Adapter 2** to **HR**.

Adapter 1


VyOS - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Name
Finance
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027C20038
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number
COM1
IRQ
4
I/O Port
0x3F8
OK
Cancel
Help

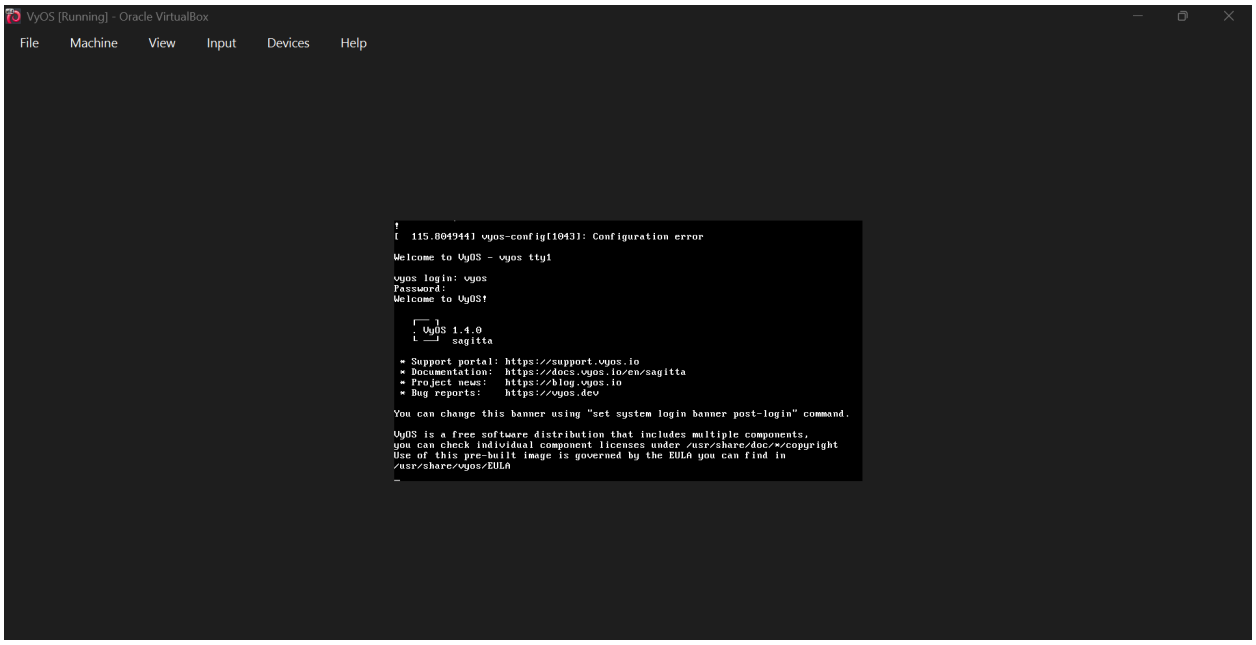
Adapter 2


VyOS - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Internal Network
Plug this virtual netw
Name
HR
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
0800272B6052
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number
COM1
IRQ
4
I/O Port
0x3F8
OK
Cancel
Help

- Start the VM.

2. Set up two interfaces to represent your network segments:

- Log into the VyOS router: **username/password:**vyos/vyos



- Access the VyOS CLI through the console: **configure.**

```
vyos@vyos:~$ configure
WARNING: There was a config error on boot: saving the configuration now could ov
erwrite data.
You may want to check and reload the boot config
[edit]
vyos@vyos#
```

- - show interfaces

```
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface    IP Address    MAC                VRF        MTU  S/L    Description
-----------  ------------  -----------------  -------  -----  -----  -----------
--
eth2         -             08:00:27:07:cd:30  default   1500  u/D
eth3         -             08:00:27:e7:a2:33  default   1500  u/D
lo           127.0.0.1/8   00:00:00:00:00:00  default  65536  u/u
             ::1/128
vyos@vyos:~$ _
```

- sudo tcpdump -i eth2 arp

```
vyos@vyos# sudo tcpdump -i eth2 arp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth2, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:56:59.528651 ARP, Request who-has 192.168.2.2 tell 192.168.2.1, length 28
06:56:59.531078 ARP, Reply 192.168.2.2 is-at 08:00:27:69:88:63 (oui Unknown), le
ngth 46
06:57:14.026634 ARP, Request who-has 192.168.2.1 tell 192.168.2.2, length 46
06:57:14.026657 ARP, Reply 192.168.2.1 is-at 08:00:27:49:f2:f4 (oui Unknown), le
ngth 28
06:57:45.609238 ARP, Request who-has 192.168.2.2 tell 192.168.2.1, length 28
06:57:45.613513 ARP, Reply 192.168.2.2 is-at 08:00:27:69:88:63 (oui Unknown), le
ngth 46
06:57:58.057654 ARP, Request who-has 192.168.2.1 tell 192.168.2.2, length 46
06:57:58.057683 ARP, Reply 192.168.2.1 is-at 08:00:27:49:f2:f4 (oui Unknown), le
ngth 28
```

- Assign eth3 to Subnet 1 (Finance):

- set interfaces ethernet eth3 address 192.168.1.1/24
- set interfaces ethernet eth3 description ‘Finance Subnet’

```
vyos@vyos# set interfaces ethernet eth3 address '192.168.1.1/24'
[edit]
vyos@vyos# set interfaces ethernet eth3 description 'Finance Subnet'
[edit]
```

Assign eth2 to Subnet 2 (HR):

- set interfaces ethernet eth2 address 192.168.2.1/24
- set interfaces ethernet eth2 description ‘HR Subnet’

```
vyos@vyos# set interfaces ethernet eth2 address '192.168.2.1/24'
[edit]
vyos@vyos# set interfaces ethernet eth2 description 'HR Subnet'
[edit]
```

- Apply and save the configuration:

- commit
- save

```
vyos@vyos# commit
save[edit]
vyos@vyos# save
[edit]
vyos@vyos#
```

3. Verify your configuration: show interfaces

```
vyos@vyos# show interfaces
 ethernet eth2 {
     address 192.168.2.1/24
     description "HR Subnet"
     hw-id 08:00:27:07:cd:30
 }
 ethernet eth3 {
     address 192.168.1.1/24
     description "Finance Subnet"
     hw-id 08:00:27:e7:a2:33
 }
 loopback lo {
 }
[edit]
vyos@vyos# _
```

4. Confirm that the interfaces are properly set with the correct IP addresses and descriptions. Mine, as shown above, is well done.
5. Verify the configurations up to this point; the HR VM should be able to ping to its gateway, and the same to Finance.

Finance gateway

```
student@finance-vm:~$ hostname -I
192.168.1.2
student@finance-vm:~$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=64 time=0.103 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=64 time=0.045 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=64 time=0.052 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=64 time=0.043 ms
64 bytes from 192.168.1.2: icmp_seq=5 ttl=64 time=0.319 ms
^C
--- 192.168.1.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4118ms
rtt min/avg/max/mdev = 0.043/0.112/0.319/0.105 ms
student@finance-vm:~$
```

HR Gateway

```
student@hr-vm:~$ hostname -I
192.168.2.2
student@hr-vm:~$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 bytes from 192.168.2.1: icmp_seq=1 ttl=64 time=3.03 ms
64 bytes from 192.168.2.1: icmp_seq=2 ttl=64 time=2.84 ms
64 bytes from 192.168.2.1: icmp_seq=3 ttl=64 time=1.86 ms
64 bytes from 192.168.2.1: icmp_seq=4 ttl=64 time=1.20 ms
64 bytes from 192.168.2.1: icmp_seq=5 ttl=64 time=1.23 ms
64 bytes from 192.168.2.1: icmp_seq=6 ttl=64 time=1.98 ms
^C
--- 192.168.2.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5006ms
rtt min/avg/max/mdev = 1.197/2.024/3.031/0.710 ms
student@hr-vm:~$
```

At this point, the Finance and HR VMs can also communicate. Here is the ping evidence below

```
student@finance-vm:~$ hostname -I
192.168.1.2
student@finance-vm:~$ ping 192.168.2.2
PING 192.168.2.2 (192.168.2.2) 56(84) bytes of data.
64 bytes from 192.168.2.2: icmp_seq=1 ttl=63 time=3.92 ms
64 bytes from 192.168.2.2: icmp_seq=2 ttl=63 time=4.95 ms
64 bytes from 192.168.2.2: icmp_seq=3 ttl=63 time=3.15 ms
64 bytes from 192.168.2.2: icmp_seq=4 ttl=63 time=3.48 ms
64 bytes from 192.168.2.2: icmp_seq=5 ttl=63 time=2.82 ms
64 bytes from 192.168.2.2: icmp_seq=6 ttl=63 time=4.19 ms
^C
--- 192.168.2.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 2.820/3.751/4.948/0.702 ms
student@finance-vm:~$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 bytes from 192.168.2.1: icmp_seq=1 ttl=64 time=1.47 ms
64 bytes from 192.168.2.1: icmp_seq=2 ttl=64 time=1.30 ms
64 bytes from 192.168.2.1: icmp_seq=3 ttl=64 time=2.78 ms
64 bytes from 192.168.2.1: icmp_seq=4 ttl=64 time=1.53 ms
64 bytes from 192.168.2.1: icmp_seq=5 ttl=64 time=1.03 ms
64 bytes from 192.168.2.1: icmp_seq=6 ttl=64 time=2.30 ms
```

But in our next steps, we want to cut off and segment this network by separating the Finance and HR.

Task 4. Firewall and Traffic Validation

1. Create firewall rules to block cross-subnet traffic:

- **Block Finance → HR traffic:**

- configure
- set firewall ipv4 name BLOCK-FINANCE-HR default-action 'accept'
- set firewall ipv4 name BLOCK-FINANCE-HR rule 10 action 'drop'
- set firewall ipv4 name BLOCK-FINANCE-HR rule 10 source address '192.168.1.0/24'
- set firewall ipv4 name BLOCK-FINANCE-HR rule 10 destination address '192.168.2.0/24'

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set firewall ipv4 name BLOCK-FINANCE-HR default-action 'accept'
[edit]
vyos@vyos# set firewall ipv4 name BLOCK-FINANCE-HR rule 10 action 'drop'
[edit]
vyos@vyos# set firewall ipv4 name BLOCK-FINANCE-HR rule 10 source address '192.1
68.1.0/24'
[edit]
vyos@vyos# set firewall ipv4 name BLOCK-FINANCE-HR rule 10 destination address '
192.168.2.0/24'
[edit]
vyos@vyos#
```

- **Block HR → Finance traffic:**

- set firewall ipv4 name BLOCK-HR-FINANCE default-action 'accept'
- set firewall ipv4 name BLOCK-HR-FINANCE rule 10 action 'drop'
- set firewall ipv4 name BLOCK-HR-FINANCE rule 10 source address '192.168.2.0/24'
- set firewall ipv4 name BLOCK-HR-FINANCE rule 10 destination address '192.168.1.0/24'

```
vyos@vyos# set firewall ipv4 name BLOCK-HR-FINANCE default-action 'accept'
[edit]
vyos@vyos# set firewall ipv4 name BLOCK-HR-FINANCE rule 10 action 'drop'
[edit]
vyos@vyos# set firewall ipv4 name BLOCK-HR-FINANCE rule 10 source address '192.1
68.2.0/24'
[edit]
vyos@vyos# set firewall ipv4 name BLOCK-HR-FINANCE rule 10 destination address '
192.168.1.0/24'
[edit]
vyos@vyos#
```

These rules explicitly drop traffic that tries to cross from one subnet into the other, but allow other traffic (such as to the router itself) by default.

2. Define Firewall Zones and Apply Policies:

Assign interfaces to zones

- set firewall zone FINANCE interface eth3
- set firewall zone HR interface eth2

```
vyos@vyos# set firewall zone FINANCE interface eth3
[edit]
vyos@vyos# set firewall zone HR interface eth2
[edit]
vyos@vyos#
```

- commit
- save

```
vyos@vyos# commit
save[edit]
vyos@vyos# save
[edit]
vyos@vyos#
```

Apply rule sets between zones

- set firewall zone HR from FINANCE firewall ipv4-name 'BLOCK-FINANCE-HR'
- set firewall zone FINANCE from HR firewall ipv4-name 'BLOCK-HR-FINANCE'

The above commands failed

set firewall zone HR default-action drop
set firewall zone FINANCE default-action drop

```
vyos@vyos# set firewall zone HR default-action drop
[edit]
vyos@vyos# set firewall zone FINANCE default-action drop
[edit]
vyos@vyos#
```

These rules assign each subnet's interface to its zone and tell VyOS which rule set governs traffic between them. The settings tell VyOS:

“If traffic originates in the Finance zone and goes toward HR, use the BLOCK-FINANCE-HR rule set.”
“If traffic originates in HR and goes toward Finance, use the BLOCK-HR-FINANCE rule set.”

3. Enable default stateful filtering:
- set firewall global-options state-policy established action 'accept'
- set firewall global-options state-policy related action 'accept'
- set firewall global-options state-policy invalid action 'drop'

4. Commit and save the changes as before:
- commit
- save

```
vyos@vyos# set firewall global-options state-policy established action 'accept'
[edit]
vyos@vyos# set firewall global-options state-policy related action 'accept'
[edit]
vyos@vyos# set firewall global-options state-policy invalid action 'drop'
[edit]
vyos@vyos# commit
save[edit]
vyos@vyos# save
[edit]
vyos@vyos# _
```

Task 5. Validate Basic Connectivity

- **Test connectivity to confirm subnet isolation.**

1. From Finance-VM, ping the Subnet 1 interface on VyOS (192.168.1.1).

```
student@finance-vm:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=1.19 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=1.35 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=1.44 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=1.31 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=64 time=1.54 ms
64 bytes from 192.168.1.1: icmp_seq=6 ttl=64 time=1.93 ms
64 bytes from 192.168.1.1: icmp_seq=7 ttl=64 time=1.89 ms
^C
--- 192.168.1.1 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6008ms
rtt min/avg/max/mdev = 1.190/1.521/1.930/0.265 ms
student@finance-vm:~$
```

2. From HR-VM, ping the Subnet 2 interface on VyOS (192.168.2.1).

```
student@hr-vm:~$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 bytes from 192.168.2.1: icmp_seq=1 ttl=64 time=1.33 ms
64 bytes from 192.168.2.1: icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from 192.168.2.1: icmp_seq=3 ttl=64 time=1.24 ms
64 bytes from 192.168.2.1: icmp_seq=4 ttl=64 time=1.43 ms
64 bytes from 192.168.2.1: icmp_seq=5 ttl=64 time=1.08 ms
64 bytes from 192.168.2.1: icmp_seq=6 ttl=64 time=0.976 ms
64 bytes from 192.168.2.1: icmp_seq=7 ttl=64 time=2.27 ms
64 bytes from 192.168.2.1: icmp_seq=8 ttl=64 time=0.813 ms
64 bytes from 192.168.2.1: icmp_seq=9 ttl=64 time=0.769 ms
64 bytes from 192.168.2.1: icmp_seq=10 ttl=64 time=1.18 ms
^C
--- 192.168.2.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
rtt min/avg/max/mdev = 0.769/1.211/2.270/0.405 ms
student@hr-vm:~$ _
```

3. Attempt to ping between Finance-VM and HR-VM. These pings should fail, confirming that the subnets are isolated.

Finance-VM IP:192.168.1.2
HR-VM IP:192.168.2.2
- ping the HR-VM from the Finance-VM
- ping 192.168.2.2

```
student@finance-vm:~$ hostname -I
192.168.1.2
student@finance-vm:~$ ping 192.168.2.2
PING 192.168.2.2 (192.168.2.2) 56(84) bytes of data.
^C
--- 192.168.2.2 ping statistics ---
265 packets transmitted, 0 received, 100% packet loss, time 270427ms

student@finance-vm:~$
```

- ping the Finance-VM from the HR-VM
- ping 192.168.1.2

```
student@hr-vm:~$ hostname -I
192.168.2.2
student@hr-vm:~$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
^C
--- 192.168.1.2 ping statistics ---
240 packets transmitted, 0 received, 100% packet loss, time 244864ms

student@hr-vm:~$ _
```

Result explanation:
- **ping traffic between the HR-VM and Finance-VM** is successfully blocked, as expected:

- **Finance-VM (192.168.1.2) to HR-VM (192.168.2.2)** → 100% packet loss.

- **HR-VM (192.168.2.2) to Finance-VM (192.168.1.2)** → 100% packet loss.

Task 6. Capture and Analyze Traffic

1. Before capturing traffic, determine which network interface is assigned to the Finance-VM:
- ip a

```
student@finance-vm:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:df:fb:90 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.2/24 brd 192.168.1.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fedf:fb90/64 scope link
       valid_lft forever preferred_lft forever
student@finance-vm:~$
```

2. Look for the interface that has the IP address 192.168.1.2/24 assigned to it.
It may be named ens3, enp0s3, or something else, depending on VirtualBox.
Take note of the correct interface name.
From the result in 1 above screenshot:
The IP address `192.168.1.2/24` is assigned to the interface **`enp0s3`**, as shown in the output:
inet 192.168.1.2/24 brd 192.168.1.255 scope global **enp0s3**

Step 2: Start Traffic Capture on the Finance-VM
1. On Finance-VM, install tcpdump if it's not already available:
- sudo apt update && sudo apt install tcpdump

```
student@finance-vm:~$ sudo apt update && sudo apt install tcpdump
[sudo] password for student:
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:2 http://ke.archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://ke.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:4 http://ke.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:2 http://ke.archive.ubuntu.com/ubuntu noble InRelease
Err:1 http://security.ubuntu.com/ubuntu noble-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Ign:3 http://ke.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://ke.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:2 http://ke.archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://ke.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://ke.archive.ubuntu.com/ubuntu noble-backports InRelease
Err:2 http://ke.archive.ubuntu.com/ubuntu noble InRelease
  Temporary failure resolving 'ke.archive.ubuntu.com'
Err:3 http://ke.archive.ubuntu.com/ubuntu noble-updates InRelease
  Temporary failure resolving 'ke.archive.ubuntu.com'
Err:4 http://ke.archive.ubuntu.com/ubuntu noble-backports InRelease
  Temporary failure resolving 'ke.archive.ubuntu.com'
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
W: Failed to fetch http://ke.archive.ubuntu.com/ubuntu/dists/noble/InRelease  Temporary failure resolving 'ke.archive.ub
W: Failed to fetch http://ke.archive.ubuntu.com/ubuntu/dists/noble-updates/InRelease  Temporary failure resolving 'ke.ar
W: Failed to fetch http://ke.archive.ubuntu.com/ubuntu/dists/noble-backports/InRelease  Temporary failure resolving 'ke.
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/noble-security/InRelease  Temporary failure resolving 'secur
W: Some index files failed to download. They have been ignored, or old ones used instead.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tcpdump is already the newest version (4.99.4-3ubuntu4.24.04.1).
tcpdump set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
student@finance-vm:~$
```

2. Run tcpdump to monitor traffic on the correct interface (replace <interface> with the name you found in Step 1): sudo tcpdump -i <interface>
(Example: If ip a showed ens3 as the interface, run sudo tcpdump -i ens3)
- Mine showed: **enp0s3**
- sudo tcpdump -i enp0s3

```
student@finance-vm:~$ sudo tcpdump -i enp0s3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
23:54:45.435439 IP 192.168.1.2.34698 > 8.8.8.8.domain: Flags [S], seq 437130375, win 64240, options [mss 1460,sackOK,TS val 2872239415 ecr 0,nop,wscale 7], leng
th 0
23:54:46.459391 IP 192.168.1.2.34698 > 8.8.8.8.domain: Flags [S], seq 437130375, win 64240, options [mss 1460,sackOK,TS val 2872240439 ecr 0,nop,wscale 7], leng
th 0
23:54:47.484383 IP 192.168.1.2.34698 > 8.8.8.8.domain: Flags [S], seq 437130375, win 64240, options [mss 1460,sackOK,TS val 2872241464 ecr 0,nop,wscale 7], leng
th 0
23:54:48.507350 IP 192.168.1.2.34698 > 8.8.8.8.domain: Flags [S], seq 437130375, win 64240, options [mss 1460,sackOK,TS val 2872242487 ecr 0,nop,wscale 7], leng
th 0
23:54:48.763632 ARP, Request who-has 192.168.1.1 tell 192.168.1.2, length 28
23:54:48.765396 ARP, Reply 192.168.1.1 is-at 08:00:27:ef:6b:5e (oui Unknown), length 46
23:54:50.555401 IP 192.168.1.2.34698 > 8.8.8.8.domain: Flags [S], seq 437130375, win 64240, options [mss 1460,sackOK,TS val 2872244535 ecr 0,nop,wscale 7], leng
th 0
23:54:50.556885 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:54:50.557588 IP 192.168.1.2.34706 > 8.8.8.8.domain: Flags [S], seq 1571294979, win 64240, options [mss 1460,sackOK,TS val 2872244537 ecr 0,nop,wscale 7,tfo
cookiereq,nop,nop], length 0
23:54:51.579503 IP 192.168.1.2.34706 > 8.8.8.8.domain: Flags [S], seq 1571294979, win 64240, options [mss 1460,sackOK,TS val 2872245559 ecr 0,nop,wscale 7], len
gth 0
23:54:52.603843 IP 192.168.1.2.34706 > 8.8.8.8.domain: Flags [S], seq 1571294979, win 64240, options [mss 1460,sackOK,TS val 2872246583 ecr 0,nop,wscale 7], len
gth 0
23:55:05.538231 IP 192.168.1.2.39484 > 8.8.8.8.domain: 35840+ PTR? 1.1.168.192.in-addr.arpa. (42)
23:55:10.094687 IP 192.168.1.2.43611 > 8.8.8.8.domain: 8808+ PTR? 8.8.8.8.in-addr.arpa. (38)
23:55:10.096528 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 74
23:55:10.096903 IP 192.168.1.2.40045 > 8.8.8.8.domain: 31479+ PTR? 2.1.168.192.in-addr.arpa. (42)
23:55:10.098864 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 78
23:55:10.100844 IP 192.168.1.2.33630 > 8.8.8.8.domain: Flags [S], seq 536136896, win 64240, options [mss 1460,sackOK,TS val 2872264080 ecr 0,nop,wscale 7,tfo  c
ookiereq,nop,nop], length 0
23:55:11.163490 IP 192.168.1.2.33630 > 8.8.8.8.domain: Flags [S], seq 536136896, win 64240, options [mss 1460,sackOK,TS val 2872265143 ecr 0,nop,wscale 7], leng
th 0
23:55:12.187655 IP 192.168.1.2.33630 > 8.8.8.8.domain: Flags [S], seq 536136896, win 64240, options [mss 1460,sackOK,TS val 2872266167 ecr 0,nop,wscale 7], leng
th 0
23:55:17.307902 IP 192.168.1.2.33630 > 8.8.8.8.domain: Flags [S], seq 536136896, win 64240, options [mss 1460,sackOK,TS val 2872271287 ecr 0,nop,wscale 7], leng
th 0
23:55:17.309651 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:55:17.310341 IP 192.168.1.2.56278 > 8.8.8.8.domain: Flags [S], seq 3038079983, win 64240, options [mss 1460,sackOK,TS val 2872271290 ecr 0,nop,wscale 7,tfo
cookiereq,nop,nop], length 0
23:55:18.331394 IP 192.168.1.2.56278 > 8.8.8.8.domain: Flags [S], seq 3038079983, win 64240, options [mss 1460,sackOK,TS val 2872272311 ecr 0,nop,wscale 7], len
gth 0
```

3. Leave this running in the terminal—you will not be able to type commands in this window while capturing traffic.

Step 3: Perform Ping Tests (From Another Machine)

1. Since tcpdump is running on Finance-VM, you need to run ping tests from a different device (HR-VM or VyOS Router):
   a. On HR-VM, try to ping Finance-VM (should be blocked): ping 192.168.1.2

```
student@hr-vm:~$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
^C
--- 192.168.1.2 ping statistics ---
303 packets transmitted, 0 received, 100% packet loss, time 309331ms

student@hr-vm:~$
```

   b. On VyOS Router, try to ping Finance-VM (should succeed): ping 192.168.1.2

```
vyos@vyos# ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=64 time=1.78 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=64 time=1.47 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=64 time=2.02 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=64 time=1.38 ms
64 bytes from 192.168.1.2: icmp_seq=5 ttl=64 time=1.90 ms
64 bytes from 192.168.1.2: icmp_seq=6 ttl=64 time=1.62 ms
64 bytes from 192.168.1.2: icmp_seq=7 ttl=64 time=1.59 ms
64 bytes from 192.168.1.2: icmp_seq=8 ttl=64 time=2.16 ms
64 bytes from 192.168.1.2: icmp_seq=9 ttl=64 time=1.25 ms
64 bytes from 192.168.1.2: icmp_seq=10 ttl=64 time=1.83 ms
64 bytes from 192.168.1.2: icmp_seq=11 ttl=64 time=1.57 ms
64 bytes from 192.168.1.2: icmp_seq=12 ttl=64 time=2.12 ms
64 bytes from 192.168.1.2: icmp_seq=13 ttl=64 time=1.69 ms
64 bytes from 192.168.1.2: icmp_seq=14 ttl=64 time=1.42 ms
^C
--- 192.168.1.2 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13033ms
rtt min/avg/max/mdev = 1.249/1.699/2.156/0.270 ms
[edit]
vyos@vyos#
```

2. Observe whether these pings appear in the tcpdump output on Finance-VM.

```
23:44:35.723485 IP 192.168.1.2.60970 > 8.8.8.8.domain: Flags [S], seq 1404672261, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:36.731993 IP 192.168.1.2.60970 > 8.8.8.8.domain: Flags [S], seq 1404672261, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:37.756418 IP 192.168.1.2.60970 > 8.8.8.8.domain: Flags [S], seq 1404672261, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:39.803450 IP 192.168.1.2.60970 > 8.8.8.8.domain: Flags [S], seq 1404672261, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:39.805517 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:44:39.806927 IP 192.168.1.2.60982 > 8.8.8.8.domain: Flags [S], seq 4079197625, win 64240, options [mss 1460,sackOK,TS
cookiereq,nop,nop], length 0
23:44:40.827404 IP 192.168.1.2.60982 > 8.8.8.8.domain: Flags [S], seq 4079197625, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:41.852384 IP 192.168.1.2.60982 > 8.8.8.8.domain: Flags [S], seq 4079197625, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:42.875564 IP 192.168.1.2.60982 > 8.8.8.8.domain: Flags [S], seq 4079197625, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:43.899426 IP 192.168.1.2.60982 > 8.8.8.8.domain: Flags [S], seq 4079197625, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:44.896790 ARP, Request who-has 192.168.1.2 tell 192.168.1.1, length 46
23:44:44.896814 ARP, Reply 192.168.1.2 is-at 08:00:27:df:fb:90 (oui Unknown), length 28
23:44:44.923459 IP 192.168.1.2.60982 > 8.8.8.8.domain: Flags [S], seq 4079197625, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:46.972421 IP 192.168.1.2.60982 > 8.8.8.8.domain: Flags [S], seq 4079197625, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:46.973726 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:44:46.974090 IP 192.168.1.2.51280 > 8.8.8.8.domain: Flags [S], seq 4032353082, win 64240, options [mss 1460,sackOK,TS
cookiereq,nop,nop], length 0
23:44:47.995501 IP 192.168.1.2.51280 > 8.8.8.8.domain: Flags [S], seq 4032353082, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:49.019897 IP 192.168.1.2.51280 > 8.8.8.8.domain: Flags [S], seq 4032353082, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:50.044314 IP 192.168.1.2.51280 > 8.8.8.8.domain: Flags [S], seq 4032353082, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:51.067550 IP 192.168.1.2.51280 > 8.8.8.8.domain: Flags [S], seq 4032353082, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:52.092513 IP 192.168.1.2.51280 > 8.8.8.8.domain: Flags [S], seq 4032353082, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:54.139390 IP 192.168.1.2.51280 > 8.8.8.8.domain: Flags [S], seq 4032353082, win 64240, options [mss 1460,sackOK,TS
gth 0
23:44:54.140562 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:44:54.141748 IP 192.168.1.2.42146 > 8.8.8.8.domain: 17178+ AAAA? ntp.ubuntu.com. (32)
23:44:54.143580 IP 192.168.1.2.55622 > 8.8.8.8.domain: 53924+ A? ntp.ubuntu.com. (32)
23:44:59.343431 IP 192.168.1.2.42146 > 8.8.8.8.domain: 17178+ AAAA? ntp.ubuntu.com. (32)
23:44:59.344975 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:44:59.345742 IP 192.168.1.2.55622 > 8.8.8.8.domain: 53924+ A? ntp.ubuntu.com. (32)
23:44:59.348562 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:44:59.349440 IP 192.168.1.2.49392 > 8.8.8.8.domain: Flags [S], seq 4052268581, win 64240, options [mss 1460,sackOK,TS
cookiereq,nop,nop], length 0
```

Step 4: Apply Filters to Focus on Specific Traffic

Instead of capturing all packets, you can filter traffic related to the VyOS router:

- sudo tcpdump -i <interface> host 192.168.1.1
- My interface is enp0s3
- sudo tcpdump -i enp0s3 host 192.168.1.1

```
student@finance-vm:~$ sudo tcpdump -i enp0s3 host 192.168.1.1
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
23:56:58.556773 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:05.725816 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:10.879828 ARP, Request who-has 192.168.1.2 tell 192.168.1.1, length 46
23:57:10.879849 ARP, Reply 192.168.1.2 is-at 08:00:27:df:fb:90 (oui Unknown), length 28
23:57:12.892817 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:18.097660 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 78
23:57:18.112633 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 78
23:57:23.387810 ARP, Request who-has 192.168.1.1 tell 192.168.1.2, length 28
23:57:23.388817 ARP, Reply 192.168.1.1 is-at 08:00:27:ef:6b:5e (oui Unknown), length 46
23:57:25.308458 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:32.476467 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:37.506880 ARP, Request who-has 192.168.1.2 tell 192.168.1.1, length 46
23:57:37.506903 ARP, Reply 192.168.1.2 is-at 08:00:27:df:fb:90 (oui Unknown), length 28
23:57:39.644875 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:44.845406 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 74
23:57:44.847956 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:50.011515 ARP, Request who-has 192.168.1.1 tell 192.168.1.2, length 28
23:57:50.012661 ARP, Reply 192.168.1.1 is-at 08:00:27:ef:6b:5e (oui Unknown), length 46
23:57:51.997881 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:57:59.165712 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:58:04.640414 ARP, Request who-has 192.168.1.2 tell 192.168.1.1, length 46
23:58:04.640437 ARP, Reply 192.168.1.2 is-at 08:00:27:df:fb:90 (oui Unknown), length 28
23:58:06.332455 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:58:11.344402 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:58:11.346758 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:58:16.635512 ARP, Request who-has 192.168.1.1 tell 192.168.1.2, length 28
23:58:16.636672 ARP, Reply 192.168.1.1 is-at 08:00:27:ef:6b:5e (oui Unknown), length 46
23:58:18.557536 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:58:25.724518 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
23:58:30.752378 ARP, Request who-has 192.168.1.2 tell 192.168.1.1, length 46
23:58:30.752406 ARP, Reply 192.168.1.2 is-at 08:00:27:df:fb:90 (oui Unknown), length 28
23:58:32.893537 IP 192.168.1.1 > 192.168.1.2: ICMP net 8.8.8.8 unreachable, length 68
```

This will only show packets traveling between Finance-VM and VyOS Router.
Useful for analyzing allowed vs. blocked traffic.

Step 5: Stop Capture and Document Observations

1. Press Ctrl + C to stop tcpdump.
2. Document observations:
   a. Do you see traffic crossing subnets?

Answer:

1. Testing Traffic Crossing Subnets (Ping from HR-VM to Finance-VM)

ping 192.168.1.2

```
student@hr-vm:~$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.

--- 192.168.1.2 ping statistics ---
120 packets transmitted, 0 received, 100% packet loss, time 121896ms
```

This resulted in **100% packet loss**, confirming that the firewall is blocking traffic between the HR and Finance subnets.

   b. Are pings from HR-VM blocked as expected?

The HR-VM ping test to the Finance-VM successfully showed that ping traffic was blocked, indicating that the firewall rules are working as expected.

c. Can you confirm Finance-VM can communicate with VyOS Router?

```
vyos@vyos# ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=64 time=2.80 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=64 time=1.48 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=64 time=2.35 ms
^C
--- 192.168.1.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2007ms
rtt min/avg/max/mdev = 1.484/2.210/2.800/0.545 ms
[edit]
vyos@vyos# _
```

Yes.
The ping was **successful**, confirming that Finance-VM can still communicate with VyOS despite the segmentation between Finance and HR subnets.

Reflection: Why This Matters in Cybersecurity

In this lesson, you applied network segmentation, a fundamental security principle that reduces the risk of lateral movement by attackers. By enforcing isolation between Finance and HR, you ensured that a compromised device in one subnet cannot easily access the other without passing through a controlled boundary.

In real-world environments, network segmentation is used to:

- Protect sensitive data from unauthorized access.
- Limit the spread of malware and insider threats.
- Improve network monitoring and anomaly detection.
- Comply with regulatory requirements such as PCI-DSS and HIPAA.

As you continue developing your cybersecurity skills, think about how network segmentation integrates with Zero Trust Architecture, Intrusion Detection Systems (IDS), and Network Access Control (NAC) to enhance overall security.

Next Steps: Advanced Scenarios & Final Assessment

To further solidify your understanding:

- Modify your firewall rules to allow controlled communication between Finance and HR for specific services (e.g., allowing Finance to access an HR database server while blocking all other traffic).
- Implement logging and monitoring using VyOS to track network traffic patterns.
- Prepare for the summative assessment, where you will troubleshoot misconfigurations in a simulated environment.

Summary

By following this walkthrough, you have:

- Configured network segmentation to isolate Finance and HR departments.
- Implemented firewall enforcement to prevent lateral movement.
- Verified connectivity and analyzed network traffic using tcpdump.

These skills are critical in real-world cybersecurity roles, where segmentation and monitoring form the backbone of secure enterprise environments. In the next lab, you will extend this implementation by adding service-specific firewall controls and further enhancing security configurations.